lee010518@gmail.com

Jay Ree

LinkedIn,GitHub,Website

0452134809

Melbourne-based final-year Software Engineering (Honours) student at Monash University, with completion expected in Nov 2026, and a recipient of the 2023 and 2024 Monash Engineering Dean's Honours (Top 3%). I taught FIT1045 as a Teaching Associate at Monash University in Semester 1, 2026, and am currently a Research Assistant under Dr Mohammed Eunus Ali. I also completed a Human Research Ethics-approved Honours thesis built around a 40-participant user study, and bring industry experience as co-founder and CTO of AIVA, a Full-Stack Developer Intern at JobSearch.Works, and an IT Analyst at Coles Group.

Experience

Teaching Associate – Monash University, Melbourne Victoria **Mar 2026 – Jul 2026**

- Delivered weekly applied sessions for *FIT1045: Introduction to Programming*, mentoring students on C++ programming fundamentals, control structures, data types, and the SplashKit library.
- Assessed portfolio submissions, providing feedback on code correctness, readability, and adherence to programming practices.
- Received a **SETU teaching score of 100 (Very High)** in S1 2026, against a faculty average of 86.

Research Assistant – Monash University, Clayton Victoria **Nov 2025 –**
Project: Dynamic Platoon Formation of Multi-Type Autonomous Vehicles for Sustainable Urban Mobility

- Contributed to modelling the coordination task as a constrained optimisation problem with multi-segment matching and point-wise capacity constraints, under Dr Mohammed Eunus Ali (with Prof. Aamir Cheema as co-author).
- Designed and implemented two matching algorithms, a Greedy Maximum-Weight Matching baseline and an Iterative Linear Assignment method using the Jonker–Volgenant solver, and evaluated them on synthetic scenarios with up to 400 passive and 80 active vehicles.
- Built an end-to-end experimental framework in **Python** (NumPy, SciPy, Pandas, Matplotlib) with parameter-sweep experiments and comparative benchmarking; refined experiments and communicated results in response to research feedback.

Honours Thesis Researcher – Monash University **Jul 2025 – Sep 2026**
Honours project: Adaptive Assessments (FIT4701 / FIT4702)

- Designed and ran a MUHREC-approved within-subjects user study with 40 participants, comparing rule-based and LLM-based adaptive mental-health questionnaires, supervised by Mr ABM Russel with co-supervision from Prof. John Grundy.
- Co-developed "Navigator", a **Flutter/Supabase** prototype implementing both adaptation methods, and led the statistical analysis (paired-sample t-tests, effect sizes); the LLM-based method scored significantly higher on Clarity, Relevance, and Quality.

Chief Technology Officer (CTO) – AIVA, Melbourne **Dec 2025 – May 2026**
Project: AI-Powered Personal Executive Assistant for Productivity & Time Management

- Led the end-to-end architecture of a full-stack AI assistant platform (Node.js/Express and Supabase/PostgreSQL backend, Flutter iOS client), integrating Google Calendar, Notion, and Google Cloud voice APIs, with an autonomous Google Gemini agent for scheduling, task prioritisation, and personalised recommendations.
- Directed technical strategy and sprint planning across frontend and backend teams, driving code reviews, API design decisions, and release cycles while coordinating cross-functional collaboration between developers and designers.

Full Stack Developer Intern – Job Search Works, Melbourne **May 2025 – Nov 2025**

- Built core features for an AI-powered job application SaaS, including resume–job matching and automated application workflows using **React**, **TypeScript**, and **Firebase**.
- Improved deployment reliability by setting up CI/CD with **GitHub Actions** and **Husky**.

IT Analyst – Hardware Co-Ordination Intern – Coles Group **Jul 2024 – Dec 2024**

- Coordinated hardware lifecycle management across multiple Coles Divisions, ensuring smooth deployments, upgrades, and resolution of hardware-software integration issues across functional teams.
- Automated manual reporting processes using VBA (Microsoft Macro), reducing processing time from 4 hours to 30 minutes, streamlining asset tracking and improving data accuracy across the team.
- Optimised vendor management and asset processes, securing cost-saving agreements and contributing to over $90,000 in operational savings through lifecycle planning.

Education

Monash University **July 2020 – Present [Expected Graduation: Nov 2026]**

- Bachelor of Engineering (Honours) in Software Engineering; GPA 3.759 / 4.000, current WAM 82.571. Final Honours classification pending.
- Working proficiency in **Python**, **Java**, **TypeScript/JavaScript**, **Tailwind CSS**, **MongoDB**, **SQL**, and **PHP** for software development; strong understanding of algorithms, data structures, and databases. Experienced with **Linux**, **Angular**, **React**, **Meteor**, **Firebase**, and **Git**.

Monash College (Foundation Year), GPA 4.0/4.0 (High Distinction) **July 2019 – June 2020**

Self-driven Project

Redesign Local Business Website (Glory Auto)

- Redesigned and deployed the website for a local mechanic workshop in Oakleigh using HTML, CSS, JavaScript, and EmailJS, reducing annual costs by up to $10k (no third-party VACC maintenance) and increasing traffic and reservation requests by over 50%.

Awards & Honors

2024 Monash Engineering Deans Honours (Top 3% of cohort) – Jun 2025
2023 Monash Engineering Deans Honours (Top 3% of cohort) – May 2024
Summer Vacation Research Scholarship, Faculty of IT – 2025
Industry-Based Learning Placement Scholarship – July 2024
Academic Commendation for Highest Grade in FIT2081: Mobile Application Development – May 2024
Engineering International Undergraduate Excellence Scholarship – Jun 2020

Certifications

AWS Certified Cloud Practitioner – Mar 2025

Referees

Dr Mohammed Eunus Ali, Senior Lecturer
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Research Assistant supervisor

Mr ABM Russel, Assistant Lecturer
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Honours project supervisor